

Medical Visualization Project Assessment Rubric

1. SimpleITK Data Preparation (15%)

1.1 Technical correctness (5%)

Level	Description
Excellent	.raw files fully correct; all required SimpleITK operations demonstrated properly (I/O, resampling, cropping, normalization, casting, etc.).
Good	Mostly correct; minor issues but results are usable and operations mostly correct.
Satisfactory	Several mistakes; incomplete SimpleITK operations or partially incorrect outputs.
Poor	Raw file generation incorrect or missing; major SimpleITK steps wrong or absent.

1.2 Tutorial clarity & structure (5%)

Level	Description
Excellent	Clear, well-structured tutorial; easy to follow; reproducible; includes examples/screenshots.
Good	Generally clear; minor missing details or inconsistencies.
Satisfactory	Understandable but disorganized or incomplete.
Poor	Hard to follow; unclear workflow; not reproducible.

1.3 Reviews & integration of feedback (5%)

Level	Description
Excellent	Detailed, constructive peer reviews + strong integration of received feedback.
Good	Useful reviews + partial integration of received feedback.

Level	Description
Satisfactory	Generic or shallow reviews; minimal integration.
Poor	Incorrect, unhelpful reviews; no integration of feedback.

2. Rendering Pipeline Corrections (45%)

2.1 Volume rendering corrections (20%)

Level	Description
Excellent	Correct ray–box intersection, Δs selection, TF application, and stable emission–absorption integrator with early termination.
Good	Mostly correct; minor issues in sampling or integrator.
Satisfactory	Functional but with visible artifacts or inconsistencies.
Poor	Incorrect or missing key parts of the pipeline.

2.2 Lighting corrections (15%)

Level	Description
Excellent	Correct normals, Lambertian shading, shadow-ray casting, entry/exit, opacity accumulation, and both visibility strategies.
Good	Mostly correct; minor shading or visibility issues.
Satisfactory	Partial implementation; noticeable errors or missing features.
Poor	Incorrect or missing lighting/shadow logic.

.3 Report – Rendering section (10%, *part of global 20%*)

Level	Description
Excellent	Clear explanation of all fixes, strong reasoning, good figures, well-organized results and analysis.
Good	Mostly clear; minor missing explanations or weaker figures.

Level	Description
Satisfactory	Incomplete explanations; limited analysis.
Poor	Unclear, incomplete, or missing documentation of corrections.

3. Silhouette Enhancement (40%)

3.1 Silhouette term computation (10%)

Level	Description
Excellent	Correct Fresnel-like term; stable; meaningful control of silhouette sharpness.
Good	Mostly correct; minor issues.
Satisfactory	Functional but unstable or poorly tuned.
Poor	Incorrect or missing computation.

3.2 Integration into rendering (10%)

Level	Description
Excellent	Silhouette response enhances readability; robust across datasets and TFs.
Good	Good integration but minor artifacts.
Satisfactory	Visible integration but inconsistent or problematic.
Poor	Poor or no integration.

3.3 Perceptual & performance analysis (10%)

Level	Description
Excellent	Strong comparison visuals; clear perceptual reasoning; solid timing/performance evaluation.
Good	Reasonable analysis; minor missing details.

Level	Description
Satisfactory	Basic observations with little depth.
Poor	Missing or incorrect analysis.

3.4 Report – Silhouette section (10%, *part of global 20%*)

Level	Description
Excellent	Clear method description, strong comparisons, well-argued analysis, good figures.
Good	Mostly clear; minor missing elements.
Satisfactory	Incomplete or weak explanations; limited figures.
Poor	Unclear, incomplete, missing key elements.

4. Global Report (20%)

(A single overall assessment of the full report, combining the rendering & silhouette sections.)

Level	Description
Excellent	Well-structured, clear writing, coherent narrative, strong figures, thorough explanations and justifications.
Good	Generally clear and well-organized; minor weaknesses.
Satisfactory	Understandable but with structural or analytical inconsistencies.
Poor	Poorly written, incomplete, unclear, or lacking justification.

5. Pass/Fail Modifiers (no grade weight)

Modifier	Pass	Fail
Video demonstration	Meets requirements (≤ 5 min, demo + explanation).	–10% penalty
GitHub activity	Regular, incremental commits.	–10% penalty